

AURA-GNC: Autonomous Vision-Based Relative Navigation & Crater Feature Tracking for Binary Asteroids

ESA OSIP Call for Ideas: Autonomous Software Experiments on Hera | Category 1 – Spacecraft Autonomy & GNC



Proposal ID:	ESA-OSIP-HERA-2026-RDAS-NAV-AURA-GNC	Target Core:	GR712RC LEON3 Core 1 (Bare-Metal @ 50 MHz)
Proposing Entity:	radixal s.r.o. (Brno, Czech Republic)	Memory Limit:	64 kB Stack 96.4 kB Static RAM (0 malloc)
Leadership Triad:	Bc. Viktor Lošák (PI), Ing. Petr Slepíka, Mgr. David Riedl	Standards:	ECSS-E-ST-40C Cat D MISRA-C:2012 Deterministic

EXECUTIVE SUMMARY & KEY IN-FLIGHT BUDGETS:

AURA-GNC is an autonomous, vision-based relative navigation and feature-tracking pipeline engineered for Core 1 bare-metal C. It performs real-time optical tracking of landmark craters on Dimorphos and Didymos, feeding a deterministic 9-state Extended Kalman Filter (EKF).

- **CPU Utilization:** 16.2% @ 50 MHz SPARC V8 (Peak WCET: 3.8 s / 1020×1020 AFC frame)
- **RAM Footprint:** 96.4 kB Static RAM | < 22.0 kB Stack (Zero dynamic allocation / No malloc)
- **Relative Range Accuracy:** < 1.8% error at 10–20 km proximity (validated against PALT ground truth)
- **Feature Tracking Rate:** Up to 40 verified crater features tracked across successive frames
- **Telemetry Emission:** PUS Science Packets (APID 0x482) containing 9-state navigation vectors.

1. Problem Statement & Deep-Space Constraints

Operating in close proximity to the irregular, low-gravity Didymos binary asteroid presents major navigation challenges. Ground round-trip communication latency (24–44 min) prevents closed-loop station-keeping. Traditional optical Center-of-Brightness (CoB) centroiding fails due to irregular shapes and phase-angle shadowing, causing navigation errors exceeding 15%. Safe proximity operations require onboard optical tracking of landmark craters.

2. Technical Solution & Algorithmic Architecture

AURA-GNC implements a 4-stage optical navigation engine:

- **Stage 1 (Tiny-FAST Corner Extractor):** High-speed integer FAST-9 corner detector extracts up to 60 landmark feature points per frame.
- **Stage 2 (Binary BRIEF Descriptor & Hamming Matcher):** Computes 256-bit binary descriptors and performs landmark matching using bitwise XOR/POPCOUNT in integer registers.
- **Stage 3 (Multi-Sensor Ingestion):** Ingests PALT laser altitude and AOCS gyro rates from the Data Pool to resolve optical scale ambiguity.
- **Stage 4 (9-State Fixed-Point EKF):** Propagates spacecraft position, relative velocity, and asteroid spin vector directly on board.

4. Technical Feasibility & In-Flight Resource Budget

Resource Parameter	Hera Allocation / Limit	AURA-GNC Consumption	Margin
CPU Clock & Execution	50 MHz SPARC V8 (Core 1)	3.8 s WCET per frame (16.2% load)	+83.8% CPU Idle
Stack Depth	64.0 kB max (at 0x40010000)	21.8 kB peak stack	+65.9% Stack Margin
Static RAM (BSS+Data)	Bounded Sandbox RAM	96.4 kB Static RAM (0 malloc)	Zero Heap Frag.
Range Accuracy	CoB Baseline: > 15% error	< 1.8% relative range error	8.3× Precision Boost
Telemetry Output	12.0 MB / session budget	96 bytes per estimation epoch	< 0.1 MB total

5. Industrial Implementation Roadmap & Leadership Triad

Proposing Entity: Established in 2016 in Brno, Czech Republic, **radixal s.r.o.** specializes in mission-critical embedded systems, safety-critical railway controls (AK Signal / SIL standards), air-gapped defense architectures (URC Systems), national transport backbones (CENDIS / MD ■R), and commercial C-based optical satellite image processing in Norway.

- **Bc. Viktor Lošák (Principal Investigator & Lead Architect):** Over 10 years of embedded systems architecture and mathematical algorithm design. Responsible for overall pipeline design and ESA interface compliance.
- **Ing. Petr Slepilka (Engineering Lead & Delivery Director):** Specialist in safety-critical C software, MISRA-C static verification, automated QEMU CI/CD test harness, and ECSS Cat D qualification.
- **Mgr. David Riedl (Executive Director & Governance):** Responsible for contract governance, institutional compliance with ESA rules, and resource allocation.

6. References & Scientific Baseline

- [1] **Carnelli, I., et al.**, „*The ESA Hera Mission: Detailed Characterization of the DART Impact Outcome*“, Advances in Space Research, 2022.
- [2] **Rublee, E., et al.**, „*ORB: An efficient alternative to SIFT or SURF*“, IEEE ICCV.
- [3] **Geller, D. K.**, „*Linear Covariance Techniques for Orbital Rendezvous and Proximity Operations*“, JGCD.